The Mood Juke Box project: classifying & retrieving music by its perceived emotional content

Alexandra L. Uitdenbogerd
RMIT University
GPO Box 2476V
Melbourne, 3001
Australia
sandrau@rmit.edu.au

Ian Kaminskyj
Monash University
Wellington Rd
Clayton, 3168
Australia
Ian.kaminskyj@eng.monash.edu.au

Abstract

Recently there has been a dramatic shift in how people obtain and listen to music. Songs are downloaded, played on PCs or transferred to portable MP3 players. This provides opportunities for developing novel and creative ways of locating or presenting music. One proposed method of finding music is to retrieve it by its perceived mood. This is of particular use to film and television directors who need to find suitable background music, but also has uses for the general population, such as for providing a romantic setting. This paper describes the background, aims, methodology and expected timeline of the Mood Juke Box project. Its principal objective is to develop robust techniques for mood-based retrieval of music. It is a collaborative research project between RMIT university and Monash university.

Introduction

Music is an important part of human culture, and with increasing on-line access to music, there will be greater need for technologies that improve our ability to find music in digital libraries.

In addition to the more usual methods of locating music via metadata such as title, composer or artist, other methods based on the *content* of the music, or other people's opinions of it, are being developed. Commercially, it is already possible to locate specific recordings via a sample recorded on a mobile phone (Wang 2003); the provision of recommendations via peer-based feedback exists in various forms (for example, user-rating-based recommendation of music pioneered by Shardanand and Maes, (1995)). A relatively new approach is to retrieve music by specifying the desired mood that is evoked by the music. For example, a film director may want to use music to evoke "agitation" during a scene, or someone may want to locate suitable music for a romantic evening. Currently this can only be achieved if music in the database has been labelled by hand by human annotators. Continuing the example, the Confutatis movement of Mozart's Requiem might be labelled "agitated".

There is little prior work on retrieving music by mood. The topic seems to have been first explored in our 2002 paper that surveyed psychology and sociology research related to music (Uitdenbogerd and van Schyndel 2002). The first researchers to build mood-based music classifiers published in 2003 (for example Liu et al. (2003)), and it has now become a regular topic within the music retrieval field. Liu et al. (2003 and 2006) described a music mood classifier that used seven audio timbre features, plus audio intensity and rhythm features to classify songs into one of four moods based on the two-dimensional (2D) model of emotions. As music often varies in mood as it progresses, the researchers also implemented a mood tracker. Experimental evaluation was based on musical experts' labelling of the mood of a collection of 250 works from the classical and romantic eras of classical music.

Feng et al. (2003) also used the 2D model for classifying music according to mood. They used a small set of features based on the tempo or pace of the music and articulation, or how the tune is played. Access to music was via representation of the pieces in a 2D mood space.

While the prior work in the field demonstrates mood classification techniques and a few user interface options for mood-based browsing, a thorough investigation remains to be achieved. In particular, the full integration of results from the field of music psychology has not yet occurred for mood-based music retrieval.

Project objectives

The aims of the Mood Juke Box project are to address the problem of music retrieval according to subjective attributes, such as the evoked mood of the music, and to discover how best to deduce the associated mood of a piece of a music through analysis of the audio signal itself. In order to address the above, the following more detailed aims will be examined.

Aim 1: *Determine the best way to represent mood classifications for retrieval of music by mood.*
There are two main ways that mood is quantified in music psychology experiments: a set of mood labels (Farnsworth 1958, Schubert 2003) (such as "cheerful", "sad", "calm", or "agitated"), and a 2D representation consisting of arousal and valence (Schubert 1999). The 2D approach as implemented in prior work (for example Liu et al. 2003 and 2006) limits the the mood categories to just four. However, there are more moods that are associated with

music. If humans can distinguish them, then a mood classifier should also do so.
Some music varies in its emotion content as it progresses (Schubert 2004). It may be necessary to classify the start and end moods of music, for example, to allow a smooth mood progression for a playlist, that is, a list of musical works that are played in succession. A related problem is that listeners may wish to avoid certain moods in the music that is playing. For this to be possible, the detection of changes in mood in the music would be required.

Aim 2: *Determine the best way to evaluate fuzzy classification problems, such as the mood of music.*
Most problems that require classification, such as the presence or absence of cancer, tend to be well-defined, and rely on what is often called a "gold standard" or "ground truth". These denote a set of values that are assumed to be true for the problem, to allow systematic classifier training and evaluation. In the case of music mood, the choice of mood label for a given piece may differ depending on the listener. In the field of information retrieval, subjective assessments of the relevance of documents are used as the ground truth for evaluating document ranking algorithms (Sparck Jones 1995). A similar approach is applied to music retrieval based on melodic content, or generally perceived similarity to a given query piece (Downie 2006). However, in the retrieval case, binary or graded relevance judgements (that is, an item is classed as relevant or irrelevant to a query) are made between document and query pairs, whereas for classification, items in a collection belong to a class independent of a query. As with multiple relevance judgements for a query-document pair, multiple assessments of music according to mood label will lead to different assigned labels.

Aim 3: *Evaluate attributes of music that are needed to predict mood.*
Psychological studies suggest that tempo, tonality, distinctiveness of rhythm and pitch height are very important predictors of mood in music. In addition, however, other aspects of a musical arrangement, such as timbre and the strictness of the timing can also affect perceived mood (Livingstone and Brown 2005, Uitdenbogerd and van Schyndel 2002). Our aim is to evaluate these attributes.

Aim 4: *Determine the most appropriate interfaces for retrieving music according to mood.*
One possible approach is to visually cluster music based on the 2D mood model. However, as music collections can be large, a method for meaningfully managing the access to the collection is not immediately obvious.

Aim 5*: Determine whether music mood can be used to improve the accuracy of music recommender systems.*
User ratings have successfully been used to predict whether someone will like a new piece of music (Shardanand and Maes 1995). In 2002, we proposed that both demographic data and data from the music itself could be incorporated to improve music recommendations (Uitdenbogerd and Zobel 2002). Since then, we have shown that demographics on their own can work as well as typical collaborative filtering algorithms (algorithms that recommend new items based on user ratings) (Yapriady and Uitdenbogerd 2005). More recently, Yoshii et al. (2006) implemented a hybrid recommender system that uses audio features in addition to user ratings. These were low-level features rather than those associated with such perceived attributes as genre and mood.
Part of this project will involve comparing the effects of combining high-level and low-level features for music recommendation.

Aim 6: *Develop efficient techniques for mood-based music retrieval.*
The methods that are best in terms of *efficiency* for mood-based music retrieval will need to be determined after *effective* retrieval techniques have been discovered. An indexing technique is likely to be needed for efficient retrieval.

Methodology

For each of the aims outlined above, the following methodology will be applied in achieving these objectives.

Aim 1: Representing Mood

Experiments will be designed that compare the use of the 2D model and the emotion-label approach for music classification according to mood. These experiments will involve user studies as well as automated classification techniques. Furthermore, user studies will examine the continuously varying mood problem.

Aim 2: Fuzzy Classification

Few approaches have been proposed for effective evaluation of fuzzily defined classification problems in the field of music (Kaminskyj 2004, Mitri, Uitdenbogerd and Ciesielski 2004). There are, however, methods that have been applied in other domains such as computational linguistics (for example Grefenstette (1993)). These will be compared in the context of music mood classification, in addition to a method we have developed based on confusion matrices. The data collected from the user studies will be used for these experiments.

An alternative approach is the use of fuzzy classifiers (Yang, Liu and Chen 2006), in which the uncertainty of the mood classes is built into the system itself. Yang et al. evaluated their fuzzy system using standard techniques for classification evaluation.

Aim 3: Finding Mood Attributes in Audio

In order to establish the audio attributes that predict mood, several approaches will be used. First, we will verify the results from the psychology literature with data that has been labeled according to the features found to be important, such as tempo and tonality. Second, low level features and existing audio extraction techniques will be tested

to see how well they model the important mood attributes. Third, existing compositional references on mood will be consulted to potentially find more attribute ideas. Fourth, new features will be developed that better model the attributes associated with mood that currently are not represented well enough by the existing features.

Classification Methodology

As shown in Figure 1, during the development, testing and subsequent operation of a pattern recognition system, three separate stages are required: data collection, data preprocessing, and classification.



Figure 1 Typical Pattern Recognition System

The first stage will involve collection of appropriate data to develop and test the classification system. Stage two relates principally to the extraction of features that will allow reliable and efficient operation of the classifier. Musical pieces need to be preprocessed so that features relevant to the classification method can be extracted.

The purpose of feature extraction is threefold: (a) to reduce the amount of data that is handled by the classifier, in comparison with the full recording, (b) to produce measurements which are invariant to extraneous variable variations (for example, the pitch range of the piece), and (c) to make the classification task as easy as possible for the classifier. Achieving (a) reduces the data storage requirements of the classifier and improves its speed of operation. Applying (b) allows improvement in classification reliability in the face of extraneous variable variations. Accomplishing (c) improves classification reliability.

Following stage two, the classifier makes use of the features extracted to arrive at a decision as to the most likely class that an unknown input sample belongs to. In this application, this relates to the most likely perceived mood of the musical piece. The amount of effort that is expended in selecting high quality features will directly determine the difficulty of the classification task. Where features have been found with very high discriminatory power for the classes being separated, the classification task becomes trivial and almost any classifier will do. If such powerful features cannot be located, the task undertaken by the classifier becomes much more difficult. For these situations, its performance becomes pivotal in determining the recognition capability of the overall classification system. Therefore, considerable effort will be directed towards devising a powerful combination of features so as to make the classification task as easy and reliable as possible.

Various classifiers will be tested, however it is possible that multiple classifiers may be more appropriate than a single one. These can be arranged in a number of different architectures. In addition, when combining the results of multiple classifiers, confusion matrices can be used with considerable success to determine the best overall decision, based on the results of the individual classifiers.

Aim 4: User Interfaces

To test interfaces for the Mood Juke Box, several different interfaces will be designed and built. User studies will then be run to determine their usability. These studies will require users to complete tasks, such as retrieve music for scenes in a film. Using standard usability methodology, user properties will be measured, including:

- learnability: the time taken to reach proficiency in the task,
- efficiency: the time taken to complete a task,
- memorability: user retention over time,
- error rate: the number of errors a user makes while carrying out a task, and
- satisfaction: the user's subjective opinion of the system

Aim 5: Recommender System

The use of mood data will be tested within a recommender system by comparing different formulations of evidence such as user ratings, demographic data and mood data, to determine the relative accuracy of recommendations to users. For example, linear sum with coefficients for scaling will be tested by learning optimal coefficient values on a training data set, and then performing evaluation on test data sets.

Aim 6: Efficiency

Once all the above research has been completed, several possible retrieval implementations will be compared for efficiency through the use of timing and space use experiments.

Timeline & Deliverables

We have commenced work on the project this year, and expect it to run until the end of 2010. During this time, several sub-phases will be defined with specific deliverables expected and managed. The breakdown of these phases is given below.

Phase 1: 2007

- Install signal processing development environment and other required software,
- Develop recommendation and user profile collection software,
- Develop software for mood-related feature extraction techniques,
- Define test bed and evaluation methodology, and
- Design core approach for Phase 2.

Phase 2: 2008

- Obtain ethics approval,
- Develop methodologies for evaluation of fuzzy classification systems,
- Define and develop a music categorisation system according to mood, and
- Collect mood information from listeners.

Phase 3: 2009-Mid 2010

- Present initial findings at conferences,
- Run experiments that compare user-preference based methods of recommendation to those incorporating mood,
- Develop techniques for presenting mood-based music selections, and
- Enhance machine listening based music categorisation system by incorporating mood.

Phase 4: Mid 2010 onwards

- Run experiments that compare user interfaces,
- Present results, and
- Prepare documentation and finalise project.

Conclusion

The increased availability of music on-line justifies the development of a variety of ways for users to interact with and retrieve it. The Mood Juke Box project aims to develop robust techniques for mood-based retrieval of music, and to verify these with a solid evaluation methodology. It is a collaborative research project between RMIT university and Monash University, which is currently in the prototype development stage.

References

Downie, J. S. 2006. "The music information retrieval evaluation exchange (MIREX)", *D-Lib Magazine*, 12(12).

Farnsworth, P. R. 1958. *The Social Psychology of Music*. Holt, Rinehart and Winston, New York.

Feng Y., Zhuang Y. and Pan. Y. 2003. "Music information retrieval by detecting mood via computational media aesthetics", *In J. Liu, editor, Proc. IEEE International Conference on Web Intelligence*, pp. 235-241, Washington, DC, USA, October. IEEE, IEEE Computer Society.

Grefenstette, G. 1993. "Evaluation techniques for automatic semantic extraction: Comparing syntactic and window-based approaches", *Technical report*, University of Pittsburgh.

Kaminskyj, I. 2004. *Automatic Recognition of Musical Instruments Using Isolated Monophonic Sounds.* PhD thesis, Electrical and Computer Systems Engineering Department, Monash University, Melbourne, Victoria, Australia.

Liu, D., Lu, L. and Zhang, H.-J. 2003. "Automatic mood detection from acoustic music data", *In H. H. Hoos and D. Bainbridge, editors, International Conference on Music Information Retrieval*, Vol. 4, pp. 81-88, Baltimore, MD, October.

Livingstone, S. R. and Brown. A. R. 2005. "Dynamic response: Real-time adaptation for music emotion", *In Y. Pisan, editor, Australasian Conference on Interactive Entertainment*, Sydney, Australia, November.

Liu, D., Lu, L. and Zhang, H.-J. 2006. "Automatic mood detection and tracking of music audio signals", *IEEE Transactions on Audio, Speech, and Language Processing*, 14(1):5-18, January.

Mitri, G. , Uitdenbogerd, A. L. and Ciesielski. V. 2004. "Automatic music classification problems", *Proc. Australasian Computer Science Conference*.

Schubert, E. 1999. "Measuring emotion continuously: Validity and reliability of the two dimensional emotion space", *Australian Journal of Psychology*, 51:154-165.

Schubert, E. 2003. "Update of Hevner's adjective checklist", *Perceptual and Motor Skills*, 96:1117-1122.

Schubert, E. 2004. "Modeling perceived emotion with continuous musical features", *Music Perception*, 21(4):561-585.

Shardanand, U. and Maes, P. 1995. "Social information filtering: Algorithms for automating word of mouth" *ACM Conference on Computer Human Interaction*, pp. 210-217. ACM Press.

Sparck Jones, K. 1995. "Reflections on TREC", *Information Processing and Management*, 31(3):291-314.

Uitdenbogerd, A. L. 2002. *Music Information Retrieval Technology*. PhD thesis, School of Computer Science and Information Technology, RMIT University, Melbourne, Victoria, Australia.

Uitdenbogerd, A. L. and van Schyndel, R. G. 2002. "A review of factors affecting music recommender success" *In M. Fingerhut, editor, Third International Conference on Music Information Retrieval*, pages 204-208, Paris, France, October.

Uitdenbogerd, A. L. and Zobel, J. 1998. "Manipulation of music for melody matching", *In B. Smith and W. Effelsberg, editors, Proc. ACM International Multimedia Conference*, pp. 235-240, Bristol, UK, September. ACM, ACM Press.

Uitdenbogerd, A. L. and Zobel, J. 2002. "Music ranking techniques evaluated", *In M. Oudshoorn, editor, Proc. Australasian Computer Science Conference*, Melbourne, Australia, January.

Wang, A. 2003. ""An industrial-strength audio search algorithm", *In H. H. Hoos and D. Bainbridge, editors, International Conference on Music Information Retrieval*, Vol. 4, pp. 81-88, Baltimore, MD, October.

Yang, Y.-H., Liu, C.-C. and Chen, H. H. 2006. "Music emotion classification: a fuzzy approach", *In Y. Rui, W. Klas, and K. Mayer-Patel, editors, Proc. ACM International Multimedia Conference,* pp. 81-84, Santa Barbara, CA, USA, October. ACM, ACM Press.

Yapriady, B. and Uitdenbogerd, A. L. 2005. "Combining demographic data with collaborative filtering for automatic music recommendation"

In R.Khosla, R.J.Howlett, and L.C.Jain, editors, Knowledge-Based and Intelligent Information and Engineering Systems, Vol. 9. KES, Springer, September.

Yoshii, K., Goto, M., Komatani, K., Ogata, T. and Okuno, H.G. 2006. "Hybrid collaborative and content-based music recommendation using probabilistic model with latent user preferences", *In Tindale A. Dannenberg R., Lemstrm K., editor, International Conference on Music Information Retrieval*, Vol. 7, October.